# ASHRAE 90.4
# Energy Standard for Data Centers


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STANDARD

ANSI/ASHRAE Standard 90.4-2019
(Supersedes ANSI/ASHRAE Standard 90.4-2016)
Includes ANSI/ASHRAE addenda listed in Appendix E

# Energy Standard for Data Centers

See Informative Appendix E for ASHRAE and ANSI approval dates.

This Standard is under continuous maintenance by a Standing Standard Project Committee (SSPC) for which the Standards Committee has established a documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the Standard. Instructions for how to submit a change can be found on the ASHRAE® website (www.ashrae.org/continuous-maintenance).

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# 1.PURPOSE

The purpose of this standard is to establish the minimum energy efficiency requirements of data centers for

- a. design, construction, and a plan for operation and maintenance
- b. use of on-site or off-site renewable energy resources.

# 2. SCOPE

**2.1** This standard applies to

- a. new data centers, or portions thereof, and their systems;
- b. new additions to data centers, or portions thereof, and their systems; and
- c. modifications to systems and equipment in existing data centers or portions thereof.

**2.2** The provisions of this standard do not apply to

- a. telephone exchanges,
- b. essential facilities, and
- c. information technology equipment (ITE).

- **4. ADMINISTRATION AND ENFORCEMENT**
- New *Buildings*
- *Additions* to *Existing Buildings*
- *Alterations* of *Existing Buildings*
- Replacement of Portions of *Existing Buildings*
- Changes in *Space* Conditioning

# 5. BUILDING ENVELOPE

- Section 5 specifies the requirements for the *building envelope*.
- **Compliance.** The *building envelope* shall comply with ANSI/ASHRAE/IES Standard 90.1, Section 5.

# 6. HEATING, VENTILATING, AND AIR CONDITIONING

- specifies the requirements for *heating, ventilating, and air-conditioning* (*HVAC) systems* installed to serve *data center spaces*. *HVAC systems* installed to serve other *spaces* shall comply with ANSI/ASHRAE/IES Standard 90.1, The *HVAC system* shall comply with

- Section 6.1, “General”;
- Section 6.4,“Mandatory Provisions,”
- Section 6.5, “Maximum *Annualized Mechanical Load Component (Annualized MLC)*,”
- Section 6.6, “Submittals.”

# 6.4 Mandatory Provisions

- **Verification of *Equipment Efficiencies*.**

*Equipment efficiency* information supplied by *manufacturers* shall be verified by one of the following:

- a. *Equipment* covered under the Energy Policy Act of 2005 (EPAct) shall comply with U.S.Department of Energy certification requirements.
- b. If a certification program exists for a piece of *equipment* and it includes provisions for verification and challenge of *equipment efficiency* ratings, then the product shall be listed in the certification program.
- c. If a certification program exists for a piece of *equipment* and it includes provisions for verification and challenge of *equipment efficiency* ratings but the product is not listed in the existing certification program, the ratings shall be verified by an independent laboratory test report.
- d. If no certification program exists for a piece of *equipment*, the *equipment efficiency* ratings shall be supported by data furnished by the *manufacturer*.
- e. Where components such as indoor or outdoor coils from

- **6.5 Maximum *Annualized Mechanical Load Component* (*Annualized MLC*).**
- *Annualized MLC* shall be calculated using Equation 6.5. The resulting value shall be less than or equal to the value in Table 6.5, "Maximum *Annualized Mechanical Load Component (Annualized MLC)*."

$$(Annualized\ Mechanical\ Load\ Component) = \frac{\begin{pmatrix} \text{Mech\_}Energy25\% + \text{Mech\_}Energy50\% + \\ \text{Mech\_}Energy75\% + \text{Mech\_}Energy100\% \end{pmatrix}}{\begin{pmatrix} \text{Data Center } ITE\ Energy25\% + \\ \text{Data Center } ITE\ Energy50\% + \\ \text{Data Center } ITE\ Energy75\% + \\ \text{Data Center } ITE\ Energy100\% \end{pmatrix}} \tag{6.5}$$

where

$$\begin{gathered} \text{Mech\_}Energy\text{X\%} = \text{Total Annual } Cooling\ Energy + \\ \text{Pump } Energy + \text{Heat Rejection Fan } Energy + \\ \text{Air-Handler Fan } Energy \end{gathered}$$

where each term is a constant value calculated at each of the following *ITE* loads: 25%, 50%, 75%, 100%.

**Table 6.5 Maximum *Annualized Mechanical Load Component* (*Annualized MLC*)**

| Climate Zones as Listed in ASHRAE Standard 169 | HVAC Maximum *Annualized MLC* for *Data Center ITE Design Power* > 300 *kW* | HVAC Maximum *Annualized MLC* for *Data Center ITE Design Power* ≤ 300 *kW* |
|---|---|---|
| 0A | 0.25 | 0.31 |
| 0B | 0.28 | 0.34 |
| 1A | 0.26 | 0.31 |
| 1B | 0.27 | 0.32 |
| 2A | 0.23 | 0.29 |
| 3A | 0.21 | 0.27 |
| 4A | 0.18 | 0.26 |
| 5A | 0.16 | 0.25 |
| 6A | 0.16 | 0.24 |
| 2B | 0.17 | 0.27 |
| 3B | 0.17 | 0.26 |
| 4B | 0.14 | 0.24 |
| 5B | 0.14 | 0.23 |
| 6B | 0.14 | 0.24 |
| 3C | 0.14 | 0.23 |
| 4C | 0.14 | 0.23 |
| 5C | 0.14 | 0.23 |
| 7 | 0.14 | 0.23 |
| 8 | 0.13 | 0.22 |

STANDARD

ANSI/ASHRAE Standard 169-2013

# Climatic Data for Building Design Standards

See Appendix D for approval dates by the ASHRAE Standards Committee, the ASHRAE Board of Directors, and the American National Standards Institute.

This standard is under continuous maintenance by a Standing Standard Project Committee (SSPC) for which the Standards Committee has established a documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. The change submittal form, instructions, and deadlines may be obtained in electronic form from the ASHRAE Web site (www.ashrae.org) or in paper form from the Manager of Standards. The latest edition of an ASHRAE Standard may be purchased from the ASHRAE Web site (www.ashrae.org) or from ASHRAE Customer Service, 1791 Tullie Circle, NE, Atlanta, GA 30329-2305. E-mail: orders@ashrae.org. Fax: 678-539-2129. Telephone: 404-636-8400 (worldwide), or toll free 1-800-527-4723 (for orders in US and Canada). For reprint permission, go to www.ashrae.org/permissions.

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Includes Web-based access to climatic data, design conditions, figures, and tables. (Requires Adobe Acrobat® and Microsoft Excel®)

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**Example:**

*Data Center ITE Energy*50% for a design *ITE* load of 1000 *kW*

*Data Center ITE Energy*50% = 1000 *kW* × 8760 hours/year × 0.5 = 4,380,000 kWh

- **6.5.1.2 Calculated Quantity of Operating Units (*N*).** As shown in Table 6.5.1.2, the number of HVAC units required to meet the load can vary based on *ambient air design conditions* or a host of other factors determined by the *design professional*. When *redundant equipment* is provided, it shall be permitted to be used in calculations to demonstrate compliance only when the design uses partially loaded *equipment efficiencies*.

Table 6.5.1.2 *Building Energy* Calculation Example—Use of *Redundant Equipment*

| Example Project's Basis of Design Intent | Example's *N* (*Equipment* Installed to Meet Design Load) | Example's *R* (*Redundant Equipment* Desired to Improve Reliability) | Total Units Installed | Method of Calculation to Show Compliance with MLC (Table 6.5) |
|---|---|---|---|---|
| If constant-volume *equipment* is to be selected at less extreme conditions (e.g., ASHRAE 0.4% climate data) | 8 | 2[a] | 10 | Calculation may be based on 8 operating units (*redundant* units might not be operating). |
| Same *data center*, except constant-volume *equipment* is to be selected based on more extreme conditions (e.g., ASHRAE 20-year extreme max. wet bulb) | 10[b] | 2[a] | 12 | Calculation may be based on only 8 operating units (because only four units were determined to be required at ASHRAE 0.4% climate data; other units might not be operating). |
| If variable-volume *equipment* is to be selected at less extreme conditions (e.g., ASHRAE 0.4% climate data) | 8 | 8[a] | 16 | If variable-speed equipment (for example VFDs or ECM) is provided for fans or pumps, MLC may be calculated based on 16 operating units, using *manufacturer's* partial-load unit *efficiencies*. |

a. The *system's energy* calculation may take credit for operating available *redundant equipment* if calculated using partially loaded *equipment efficiencies*.

b. Ten (10) units, because the more severe outdoor conditions require a derate of the selected units, thereby requiring more units to meet the *N* requirement.

# 6.6 Submittals

## Completion Requirements:

- **Drawings**
- *Record drawings* shall include, as a minimum, the location and performance data on each piece of *equipment*; general configuration of the duct and pipe *distribution system*, including sizes; and the *terminal* air or water design flow rates.

## Completion Requirements (continue):

**Manuals** (operating manual and a maintenance manual)

- a. Submittal data stating *equipment* size and selected options for each piece of *equipment* requiring maintenance.
- b. Operation manuals and maintenance manuals for each piece of *equipment* and *system* requiring maintenance, except *equipment* not furnished as part of the project. Required routine maintenance actions shall be clearly identified.
- c. Names and addresses of at least one *service* agency
- d. HVAC *controls system* maintenance and calibration information, including wiring diagrams, schematics, and *control* sequence descriptions. Desired or field-determined set points shall be permanently recorded on *control* drawings at *control* devices or, for digital *control systems*, in programming comments.
- e. A complete narrative of how each *system* is intended to operate, including suggested set points.

# 7. SERVICE WATER HEATING

- The *service water heating* shall comply with ANSI/ASHRAE/IES Standard 90.1, Section 7

- **8. POWER**
- specifies the requirements for the building electrical systems delivering power to the data center space's IT load, and to equipment described

(Electrical *systems* delivering power for other uses and other *spaces* shall comply with ANSI/ASHRAE/IES Standard 90.1, Section 8.)

- **8.2.1 Compliance:**
- 8.4, “Mandatory Provisions”; and Section 8.7, “Submittals”; and either
- a. Section 8.5, “Maximum *Design Electrical Loss Component* (*Design ELC*) Option for Designs Involving Electrical *Systems* Only”, or
- b. Section 8.6, “Maximum *Design Electrical Loss Component* (*Design ELC*) Option for Designs Involving Both Electrical and Mechanical *Systems*.”

- **Electrical *Distribution Systems* for Mechanical Loads.** The electrical *distribution systems* serving mechanical loads shall be designed with pathway *losses* not exceeding 2%;

- **8.4.1.1 Multiple Electrical Paths.** Where there are multiple paths for any segment of the electrical *distribution system*, the calculations shall use the paths with the highest *losses* and/or lowest *efficiencies* for each segment to demonstrate compliance.

- **8.4.1.2 Minimum *Efficiency* or Maximum *Loss*.** The *design ELC* calculations shall use the minimum operating *efficiency* or maximum operating *loss* of each component unless a specific mode of operation (with higher *efficiency* or lower *loss*) is designated on the approved design documents

- **8.4.1.3 Corrections Allowed.**
- **8.4.1.4 *Incoming Electrical Service Segment.***
- **8.4.1.5 *UPS Segment Efficiency*.**
- **8.4.1.6 *ITE Distribution Segment Efficiency***
- **8.4.1.7 Combined *UPS* and Pathway *Loss* Calculations**
- **8.4.1.8 Alternate Designs**
- **8.4.1.9 Derivation of Electrical Component *Efficiencies (*a. Rated *equipment*. b. Unrated *equipment)***
- **8.4.1.10 Constant *ITE* Power.**

- **8.5 Maximum *Design Electrical Loss Component* (*Design ELC*) for Designs Involving Electrical *Systems* Only.**
- **8.6 Maximum *Design Electrical Loss Component* (*Design ELC*) for Designs Involving Both Electrical and Mechanical *Systems***

**Table 8.5 Maximum *Design Electrical Loss Component* (*Design ELC*) and ELC Segments *Systems* (IT Design Load <100 *kW*) [a]**

| *UPS Redundancy* Configuration | Single-Feed *UPS* (*N*, *N+1*, etc.) or No *UPS* [b] | | Active Dual-Feed *UPS* (*2N*, *2N+1*, etc.) [c] | |
|---|---|---|---|---|
| Calculation Percentage | 100% of IT design load segment ELC | 50% of IT design load segment ELC | 50% of IT design load segment ELC | 25% of IT design load segment ELC |
| Segments of ELC and Overall ELC | *Loss / efficiency* | *Loss / efficiency* | *Loss / efficiency* | *Loss / efficiency* |
| ***Incoming Electrical Service Segment*** | 15.0% / 85.0% | 11.0% / 89.0% | 11.0% / 89.0% | 10.0% / 90.0% |
| ***UPS Segment*** | 8.0% / 92.0% | 10.0% / 90.0% | 10.0% / 90.0% | 13.5% / 86.5% |
| ***ITE Distribution Segment*** | 6.0% / 94.0% | 4.0% / 96.0% | 4.0% / 96.0% | 3.0% / 97.0% |
| Electrical ***Loss / Efficiency*** Total | 26.5% / 73.5% | 23.1% / 76.9% | 23.1% / 76.9% | 24.5% / 75.5% |
| ELC | 0.265 | 0.231 | 0.231 | 0.245 |

**Table 8.6 Maximum *Design Electrical Loss Component* (*Design ELC*) and ELC Segments *Systems* (IT Design Load ≥ 100 *kW*)[a]**

| ***UPS Redundancy*** **Configuration** | **Single-Feed *UPS* (*N*, *N+1*, etc.) or No *UPS*[b]** | | **Active Dual-Feed *UPS* (*2N*, *2N+1*, etc.)[c]** | |
|---|---|---|---|---|
| **Calculation Percentage** | 100% of IT design load segment ELC | 50% of IT design load segment ELC | 50% of IT design load segment ELC | 25% of IT design load segment ELC |
| **Segments of ELC and Overall ELC** | *Loss / efficiency* | *Loss / efficiency* | *Loss / efficiency* | *Loss / efficiency* |
| ***Incoming Electrical Service Segment*** | 15.0% / 85.0% | 11.0% / 89.0% | 11.0% / 89.0% | 10.0% / 90.0% |
| ***UPS Segment*** | 6.5% / 93.5% | 8.0% / 92.0% | 8.0% / 92.0% | 11.0% / 89.0% |
| ***ITE Distribution System*** | 5.0% / 95.0% | 4.0% / 96.0% | 4.0% / 96.0% | 3.0% / 97.0% |
| **Electrical *Loss / Efficiency* Total** | 24.5% / 75.5% | 18.9% / 81.1% | 18.9% / 81.1% | 22.3%/77.7% |
| **ELC** | 0.245 | 0.189 | 0.189 | 0.223 |

- **8.7 Submittals**
- **8.7.1 Drawings.** *Construction documents* shall require that within 30 days after the date of *system* acceptance, *record drawings* of the actual installation shall be provided to the *building* owner, including the following:
- a. A *single-line diagram* of the *building* electrical *distribution system*
- d. Floor plans indicating locations of and areas served by all distribution
- e. All conditions used for the Basis of Design and calculations such as *UPS N+1* and *UPS economy mode* operation
- f. *Design ELC* calculations showing the actual numbers used and demonstrating compliance with the applicable Table 8.5 or Table 8.6 values

- **8.7.2 Manuals.** *Construction documents* shall require that an operating manual and maintenance
- manual be provided to the *building* owner. The manuals shall include, at a minimum, the following:
- a. Submittal data stating *equipment* rating and selected options for each piece of *equipment* requiring maintenance
- b. Operation and maintenance manuals for each piece of *equipment* requiring maintenance (Required routine maintenance actions shall be clearly identified.)
- c. Names and addresses of at least one qualified *service* agency
- d. A complete narrative of how each *system* is intended to operate

- **9. LIGHTING**
- **9.1 General**
- **9.1.1 Scope.** Section 9 specifies the requirements for the lighting.
- **9.2 Compliance Paths**
- **9.2.1 Compliance.** The lighting shall comply with ANSI/ASHRAE/IES Standard 90.1, Section
- 9.

- **10. OTHER EQUIPMENT**
- Other *equipment* shall comply with ANSI/ASHRAE/IES Standard 90.1, Section 10.

# 11. ALTERNATIVE COMPLIANCE METHOD

- **11.1.1 Sections 6 and 8 Trade-Off Method Scope.**
- **11.1.2 Sections 6 and 8 Trade-Off Method Rationale**
- **11.1.3 Trade-Offs Limited to *Building* Permit.**

**Table B-1 ASHRAE Standard 90.4 Compliance Checklist: Section 5, "Building Envelope"; Section 7, "Service Water Heating"; Section 9, "Lighting"; and Section 10, "Other Equipment"**

| | | | | |
|---|---|---|---|---|
| **Project Name:** | | | | |
| **Project Address:** | | | | |
| **Designer of Record:** | | **Email:** | | **Date:** |
| **Contact Person:** | | **Email:** | | **Phone:** |
| **City:** | | **Climate Zone:** | | **Phone:** |
| **Note:** Sections 5, 7, 9, and 10 of Standard 90.4 require compliance with the corresponding Sections 5, 7, 9, and 10 of Standard 90.1. Consequently, this form directs the user of Standard 90.4 to the Standard 90.1 compliance forms for those sections, which are included with the Standard 90.1 User's Manual. | | | | |
| **SECTION 5—BUILDING ENVELOPE** | | | | |
| **Section** | **Description** | **Pass / Fail** | **Target Value** | **Design Value** |
| **5.2** | **Compliance Paths** | | | |
| 5.2.1 | The *building envelope* complies with Standard 90.1. | | | |
| | Attach the *building envelope* compliance form for Section 5 of Standard 90.1. | | | |
| **SECTION 7—SERVICE WATER HEATING** | | | | |
| **Section** | **Description** | **Pass / Fail** | **Target Value** | **Design Value** |
| **7.2** | **Compliance Paths** | | | |
| 7.2.1 | The *service water heating* complies with Standard 90.1. | | | |
| | Attach the *service water heating* compliance form for Section 7 of Standard 90.1. | | | |
| **SECTION 9—LIGHTING** | | | | |
| **Section** | **Description** | **Pass / Fail** | **Target Value** | **Design Value** |
| **9.2** | **Compliance Paths** | | | |
| 9.2.1 | The lighting complies with Standard 90.1. | | | |
| | Attach the lighting compliance form for Section 9 of Standard 90.1. | | | |
| **SECTION 10—OTHER EQUIPMENT** | | | | |
| **Section** | **Description** | **Pass / Fail** | **Target Value** | **Design Value** |
| **10.2** | **Compliance Paths** | | | |
| 10.2.1 | The other equipment complies with Standard 90.1. | | | |

**Table B-2 ASHRAE Standard 90.4 Compliance Checklist: Section 6, "Heating, Ventilating, and Air Conditioning"**

| | | | | |
|---|---|---|---|---|
| **Project Name:** | | | | |
| **Project Address:** | | | | |
| **Designer of Record:** | | **Email:** | | **Date:** |
| **Contact Person:** | | **Email:** | | **Phone:** |
| **City:** | | **Climate Zone:** | | **Phone:** |
| **SECTION 6—HEATING, VENTILATING, AND AIR CONDITIONING** | | | | |
| **Section** | **Description** | **Pass / Fail** | **Target Value** | **Design Value** |
| **6.1** | **General** | | | |
| 6.1.1 | *HVAC systems* installed to *serve data center spaces* comply with Standard 90.4. | | | |
| | *HVAC systems* installed to serve other *spaces* comply with Standard 90.1. (Insert "NA" if there are no *spaces* other than *data center spaces*.) | | | |
| **6.4** | **Mandatory Provisions** | | | |
| 6.4.1 | *Equipment* complies with minimum *efficiencies*. | | | |
| **6.6** | **Maximum *Annualized Mechanical Load Component (Annualized MLC)*** | | | |
| | Enter *data center* design *ITE* load at 100%. | | | |
| | Enter calculated mechanical energy at 25% of the design *ITE* load. | | | |
| | Enter calculated mechanical energy at 50% of the design *ITE* load. | | | |
| | Enter calculated mechanical energy at 75% of the design *ITE* load. | | | |
| | Enter calculated mechanical energy at 100% of the design *ITE* load. | | | |
| | Enter HVAC *annualized Mechanical Load Component* (*annualized MLC*). | | | |
| | Attach calculations and other documentation to support values entered. | | | |
| | IF design does not comply with *annualized MLC*,<br>• either redesign to achieve compliance or<br>• check here AND also submit Section 11, "Alternative Compliance Method," form (Table B-4). | | | |
| **6.7** | **Submittals** | | | |
| 6.7.2.1 | *Construction documents* require *record drawings* to be submitted to the *building* owner within 90 days of *system* acceptance. | | | |
| 6.7.2.2 | *Construction documents* require operation and maintenance manual to be submitted to the *building* owner within 90 days of *system* acceptance. | | | |

**Table B-3 ASHRAE Standard 90.4 Compliance Checklist: Section 8, "Power"**

| | | |
|---|---|---|
| **Project Name:** | | |
| **Project Address:** | | |
| **Designer of Record:** | **Email:** | **Date:** |
| **Contact Person:** | **Email:** | **Phone:** |
| **City:** | **Climate Zone:** | **Phone:** |

**SECTION 8—POWER**

| Section | Description | Pass / Fail | Target Value | Design Value |
|---|---|---|---|---|
| **8.1** | **General** | | | |
| 8.1.1 | Electrical *systems* delivering power to *data center space's ITE* load comply with Standard 90.4. | | | |
| | Electrical *systems* delivering power for other uses and other *spaces* comply with Standard 90.1. (Insert "NA" if there are no *spaces* other than *data center spaces*.) | | | |
| **8.4** | **Mandatory Provisions** | | | |
| 8.4.1 | Electrical *systems* serving mechanical *systems* have pathway *losses* not exceeding 2%. | | | |
| **8.5** | **Maximum *Design Electrical Loss Component* (*Design ELC*) Option for Designs Involving Electrical Systems Only** | | | |
| | Enter *data center* design *ITE* load at 100%. | | | |
| | **Single-Feed *UPS* (*N*, *N+1*, etc.) or No *UPS*** | | ***Loss / Efficiency*** | ***Loss / Efficiency*** |
| | Enter incoming electrical service segment at 100% of *ITE* design load segment ELC. | | / | / |
| | Enter *UPS segment* at 100% of *ITE* design load segment ELC. | | / | / |
| | Enter *ITE* distribution system at 100% of *ITE* design load segment ELC. | | / | / |
| | Enter electrical *loss/efficiency* total at 100% of *ITE* design load segment ELC. | | / | / |
| | Enter ELC at 100% of *ITE* design load segment ELC. | | / | / |
| | Enter incoming electrical *service* segment at 50% of *ITE* design load segment ELC. | | / | / |
| | Enter *UPS segment* at 50% of *ITE* design load segment ELC. | | / | / |

# TC 9.9 Data Center Environmental Guidelines Incorporated in EU Regulation

- This works on design and operation of energy efficient data centers and IT equipment.
- TC 9.9 was requested to review and provide input on an European Union (EU) Regulation for servers and data storage products.
- Ecodesign Regulation for servers and data storage products was approved at the September 2018
- Portions of the regulation will take effect on March 1, 2020 with the final items being implemented on Jan. 1, 2023.

# Three main areas of focus on IT equipment design:

- Minimum PSU (power supply unit) efficiency and power factor requirements
- Manufacturers requirements that ensure that joining, fastening or sealing techniques do not prevent the disassembly for repair or reuse purposes
- Idle state power and active power efficiency requirements for servers with one or two processor sockets and for some of its key components; and a number of data sets that manufacturers will need to provide to verify compliance

# Thank you

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